Answer Key

Grade 6 Mathematics · Unit 6: Expressions and Operations · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 6-1

Dividing by a Unit Fraction · Dividing a Fraction by a Whole Number

Sheet 1 · Dividing by a Unit Fraction

1. 12 (each whole has 3 thirds; 4 × 3 = 12)
2. a. 12 b. 15 c. 32 d. 18 e. 21 f. 60 g. 35 h. 36
3. 6 ÷ 1/4 = 6 × 4 = 24 bags
4. 5 ÷ 1/6 = 30 pieces; 30 − 18 = 12 pieces left
5. She found half of 8 (8 ÷ 2), not how many halves are in 8. Each whole has 2 halves, so 8 ÷ 1/2 = 8 × 2 = 16.

Sheet 2 · Dividing a Fraction by a Whole Number

1. a. 1/6 b. 2/5 c. 1/8 d. 1/9 e. 3/7 f. 1/12
2. a. 1/2 ÷ 4 = 1/8 lb b. 5/6 ÷ 5 = 1/6 L
3. 3/5 ÷ 6 = 3/5 × 1/6 = 3/30 = 1/10 gallon
4. 2/3 ÷ 4 = 2/3 × 1/4 = 2/12 = 1/6 pound
5. less than 5/8 (5/8 ÷ 4 = 5/32; sharing into 4 parts makes each part smaller)

TEST PRACTICE

6. B
7. A

LESSON 6-2

Dividing a Fraction by a Fraction · Dividing with Mixed Numbers

Sheet 1 · Dividing a Fraction by a Fraction

1. 4/5 ÷ 1/10 = 8
2. a. 6 b. 3 c. 5/2 = 2 1/2 d. 7/4 = 1 3/4 e. 2/3 f. 2
3. 1/2 ÷ 2/3 = 1/2 × 3/2 = 3/4 of the recipe
4. 5/6 ÷ 5/12 = 5/6 × 12/5 = 60/30 = 2 bookmarks
5. She did not flip the divisor (she multiplied by 2/3 instead of its reciprocal 3/2). Correct: 4/5 × 3/2 = 12/10 = 6/5 = 1 1/5.

Sheet 2 · Dividing with Mixed Numbers

1. a. 6 b. 4 c. 1 1/2 d. 4 1/2 e. 5 f. 3/4
2. a. 17/5 b. 29/6
3. 8 3/4 ÷ 1 1/2 = 35/4 × 2/3 = 35/6 = 5 5/6, so 5 full bowls; 5 × 1 1/2 = 7 1/2, and 8 3/4 − 7 1/2 = 1 1/4 cups left
4. 7 1/2 ÷ 1 1/4 = 15/2 × 4/5 = 60/10 = 6 hours
5. less than 6 1/2 (the divisor is greater than 1; 6 1/2 ÷ 1 1/4 = 5 1/5)

TEST PRACTICE

6. B
7. C

LESSON 6-3

Writing Repeated Factors as Powers · Finding the Value of a Power

Sheet 1 · Writing Repeated Factors as Powers

1. a. 4^3 b. 9^2 c. 2^7 d. 10^4 e. $(1/3)^3$ f. 0.5^2 g. 6^5 h. $(2/5)^2$
2. Row 1: base 8, exponent 3, 8 × 8 × 8. Row 2: 7^4, 7 × 7 × 7 × 7. Row 3: 11^2, base 11, exponent 2.
3. 3^5
4. He switched the base and exponent. The repeated factor 4 is the base, and it is used 5 times, so the power is 4^5.

Sheet 2 · Finding the Value of a Power

1. a. 49 b. 81 c. 125 d. 100,000 e. 64 f. 1
2. a. 1/8 b. 0.09
3. a. 3^3 b. 10^4
4. 7 × 7 × 7 = 343 in^3
5. A and C: $2^6 = 64$ and $4^3 = 64$ ($6^2 = 36$, $3^4 = 81$)

TEST PRACTICE

6. C
7. A

LESSON 6-4

Writing Numerical Expressions · Order of Operations with Powers

Sheet 1 · Writing Numerical Expressions

1. a. 7 × 8 − 10 b. 50 − 12 c. 36 ÷ 9 + 2 d. 2 × (15 − 6) e. 3 × 5 + 9 f. 3 × (42 ÷ 7)
2. 5 × 6 + 4 × 3 = 30 + 12 = $42
3. 12 × 2 + 8 × 5 − 15 = 24 + 40 − 15 = $49
4. (8 + 2) × 5
5. Sample: 4 friends each buy a $6 sandwich and a $3 drink. Total cost 4 × 9 = $36.

Sheet 2 · Order of Operations with Powers

1. a. 26 b. 50 c. 17 d. 6 e. 20 f. 24
2. a. (2 + 3) × 4 = 20 b. 24 ÷ (4 + 2) = 4
3. 3 × (64 − 4) = 3 × 60 = $180
4. 5 × 16 − 30 = 80 − 30 = 50 beads
5. Lou: add and subtract left to right, 18 − 6 = 12, 12 + 2 = 14 (Kim added first).

TEST PRACTICE

6. A
7. B

LESSON 6-5

Writing Algebraic Expressions · Evaluating Algebraic Expressions

Sheet 1 · Writing Algebraic Expressions

1. a. n + 9 b. y − 8 c. 6k d. m ÷ 5 (or m/5) e. 2x + 3 f. 7(a + 2)
2. 15L + 20
3. a. p ÷ 4 b. 50 − s
4. 11p + 5; when p = 4: 11 × 4 + 5 = $49

Sheet 2 · Evaluating Algebraic Expressions

1. a. 30 b. 23 c. 22 d. 40 e. 12 f. 10
2. a. 3 × 4 + 2 × 5 = 22 b. 16 − 5 = 11
3. $13, $29, $53
4. 100 − 6 × 5 = 100 − 30 = 70%
5. 8 (multiply first: 3 × 4 = 12, then 20 − 12 = 8)

5. He put the variable with the one-time fee. The $7 repeats for each ticket, so it goes with n: 7n + 3.

TEST PRACTICE

6. C

7. D

LESSON 6-6

Simplifying to Show Equivalence · Testing and Justifying Equivalence

Sheet 1 · Simplifying to Show Equivalence

1. a. 9a b. 6y c. 6x + 2 d. 3n + 11 e. 5b + 7 f. 7p + 9q g. 5m + 11 h. 11z

2. 2x + (x + 4) + (x + 1) = 4x + 5 cm

3. No. 3w + 3w = 6w, not $6w^2$ (for w = 2: 12 vs. 24).

4. 7x + 3 (combine 5x + 2x = 7x; the 3 has no x)

5. She combined the constant 4 with the x-terms (7 + 4 − 2 = 9). Only like terms combine: 7x − 2x = 5x, so the answer is 5x + 4.

Sheet 2 · Testing and Justifying Equivalence

1. a. 27 b. 27 c. 19. 3(x + 4) and 3x + 12 might be equivalent (the distributive property shows they are).

2. a. E b. N c. N d. E

3. Sample: n = 3 gives $n^2 = 9$ but 2n = 6, so they are not equivalent.

4. x = 4: 3x = 12 but x + 6 = 10, so they are not equivalent (one test value that gives different results is enough).

5. Distributive

TEST PRACTICE

6. B

7. D

LESSON 6-7

Factors and Greatest Common Factor · Multiples and Least Common Multiple

Sheet 1 · Factors and Greatest Common Factor

1. a. 1, 2, 3, 5, 6, 10, 15, 30 b. 1, 2, 4, 7, 14, 28 c. 1, 3, 5, 9, 15, 45 d. 1, 17 e. 1, 2, 3, 4, 6, 8, 12, 16, 24, 48 f. 1, 2, 3, 6, 7, 14, 21, 42

2. a. 4 b. 9 c. 8 d. 1 e. 12 f. 15

3. a. $2^2 \times 3^2$ b. 2×5^2 c. $2^3 \times 3^2$

4. GCF(36, 54) = 18 bags; each bag has 2 granola bars and 3 juice boxes

5. 1 is a factor of every whole number, so 15 and 28 share the factor 1. The GCF is 1.

Sheet 2 · Multiples and Least Common Multiple

1. a. 21 b. 20 c. 24 d. 15 e. 36 f. 50

2. LCM(8, 10) = 40; 5 packs of hot dogs and 4 packs of buns

3. LCM(6, 9) = 18 seconds

4. LCM of 4 and 6 (12 minutes)

5. No. LCM(4, 5) = 20 = 4 × 5, but LCM(4, 6) = 12, not 24. The product is a common multiple, but not always the least (when the numbers share a factor greater than 1).

TEST PRACTICE

6. B

7. B

LESSON 6-8

Using Properties to Rewrite Expressions · Expanding and Factoring Expressions

Sheet 1 · Using Properties to Rewrite Expressions

1. a. 6n + 12 b. 12x + 3 c. 10 + 5y d. 7a − 21 e. 6b + 10 f. 10m − 10

2. a. 6(30 + 4) = 180 + 24 = 204 b. 8(50 + 2) = 400 + 16 = 416

3. a. 9c + 12 b. 3x + 13 c. 7m + 10 d. 5t + 18

4. 6x + 24; when x = 5: 6 × 5 + 24 = 54 ft² (check: 6 × 9 = 54)

5. She did not multiply the 4 by 2. 2(3x + 4) = 6x + 8, so 6x + 8 + x = 7x + 8.

Sheet 2 · Expanding and Factoring Expressions

1. a. 15x + 6 b. 8y + 56 c. 12 + 8n

2. a. 3(2x + 5) b. 4(5y + 8) c. 7(2a + 7) d. 9(4 + 3k) e. 20(2m + 3) f. 11(b + 3)

3. 8x + 56 = 8(x + 7), so the length is x + 7 units

4. Sample: 2(6x + 12), 3(4x + 8), 4(3x + 6), or 6(2x + 4). GCF form: 12(x + 2).

5. No. 6n and 9 still share a factor of 3. The GCF is 15: 15(2n + 3).

TEST PRACTICE

6. B

7. C